

① Design and implement security group rules that allow only HTTP traffic to a web server while denying all other inbound connections in a cloud environment. Provide a justification for your configuration choices.

Ans:- Security group configuration

• Inbound Rules :-

Type	Protocol	Port	Source	Purpose
HTTP	TCP	80	0.0.0.0/0	allow web traffic
ALL	ALL	ALL	DENY (Default)	Block other Traffic

• Outbound Rules :-

Type	Protocol	Port	Destination	Purpose
ALL	ALL	ALL	0.0.0.0/0	allow response & updates

→ Implementation steps :-

- ① Go to EC2 dashboard
- ② Create a security group
- ③ Add inbound rule :-
Type: HTTP
Port: 80
Source: Anywhere
- ④ Remove or avoid adding other inbound rules.
- ⑤ Attach this security group to the web server instance.
- ⑥ Launch & test.

→ Justification :-

1> only HTTP traffic is allowed → reduced attack surface

2> All other ports are blocked

3> Follows principle of least privilege

2. Illustrate the implementation of host and application security controls to protect a cloud application handling sensitive customer data.

Ans) Host level controls:-

- > enables firewalls (only required ports open)
- > use strong authentication SSH keys instead of password
- > Regular OS patching & updates.
- > Install antivirus / intrusion detection system.

Application - level controls:-

- > uses HTTP (SSL/TLS encryption) for secure data transfer
- > Implement input validation to prevent SQL injection
- > Apply authentication & authorization.
- > uses secure APIs and token based access.

Result :- Protects sensitive customer data from unauthorized access and attacks.

3. Examine the security group configurations in a cloud environment where unauthorized access occurred and pinpoint potential vulnerabilities that could be exploited.

Ans) Common misconfiguration leading to unauthorized access:-

- Open ports like:

SSH (22) RDP open to 0.0.0.0/0

- Allowing all traffic (0.0.0.0/0) instead of restricted IPs.
- No separation between frontend and backend services.

• Missing logging & monitoring.

➤ Potential Exploits:-

- Brute-force login
- Unauthorized remote access, data breaches.

Conclusion:- Poorly configured rules increase risk of attacks.

4.) How a VPC architecture using suitable subnets can isolate web servers from databases servers in a startup environment.

Ans) A VPC design includes:-

- Public subnet:
 - ↳ web servers (accessible from internet)
- Private subnet:-
 - ↳ Database servers (no direct internet access)

Flow:-

Users → web servers → Databases
 ↓ ↓
 (Public) (Public)

Benefits:-

- Databases is protected from direct external attacks.
- Improves overall system security.

Security Measures:-

- Uses NAT Gateway for private subnet outbound access.

- Uses security groups to allow only web server traffic to access the DB.
- Apply network ACL's for additional filtering.

5) Interpret how Auto scaling ensures consistent performance during periods of high application demand.

Ans) Auto scaling ensures that applications can handle varying levels of traffic efficiently:-

• Working mechanism:-

- 1) Monitors metrics such as CPU usage, memory or request count.
- 2) Automatically adds instances when demand increases.
- 3) Removes instances when demand decreases.

Benefits:-

- Maintains consistent application performance even during traffic spikes.
- Prevent server overload and downTime.
- Improves cost efficiency by using resources only when needed.
- Provides high availability by distributing load across multiple instances.

Ex:- During a sale or event, user traffic increases rapidly. Auto scaling automatically launches additional servers to handle the load, ensuring smooth user experience.